

Vutshila Mashimbyi

BSc Computer Science

☎ 083 239 0996 ✉ vutshilamash14@gmail.com 🌐 github.com/ARTMashimbyi
in linkedin.com/in/vutshila-mashimbyi
🌐 artmashimbyi.github.io/Portfolio

About Me

Final year Computer Science student at the University of the Witwatersrand.

Passionate about anything coding, i.e: full-stack development, data science, AI/ML, and cybersecurity. Combining analytical thinking with creative problem solving to build innovative digital solutions. I bring a unique blend of technical expertise and leadership experience to every project. My journey through Wits' rigorous curriculum has equipped me with deep knowledge in algorithms, data structures, software engineering, and modern development practices.

Education

BSc Computer Science

University of the Witwatersrand

- Final year student with comprehensive education in algorithms, data structures, software engineering (Design), and database management
- Top achiever in first year residence academic rankings
- Participant in Wits Maths Competition

Certificate in Full Stack Development

FNB App Academy (2025)

- Earned 32 credits in intensive full-stack development program
- Key coursework: App Strategies, GitHub Collaboration, SDLC with HTML
- UX Design Principles, API Development, Data Management
- Business Development, Backend Development (API Building)
- AI in Development, App Marketing Strategies
- Certificate Number: A0TYA07Z025 (Attached)

Professional Experience

Computer Science Tutor

University of the Witwatersrand

- Tutored first and second year students in programming fundamentals, data structures, and algorithms
- Conducted tutorial sessions and assisted with assignments
- Helped students develop computational problem-solving skills

Scrum Master & Team Lead (Skills)

Constitutional Archive Project

- Led development team in creating full-stack constitutional archive management system
- Implemented Agile methodologies and managed sprints
- Coordinated between stakeholders and developers

Maths & Science Ambassador

Winfred Nomzamo Mandela High School

- Facilitated weekly quizzes to promote peer learning
- Recognized and mentored top academic performers

Technical Skills and Interests

- **Full Stack Development** : Certified in end-to-end app development (FNB App Academy)
- **Computer Science** : Algorithms, Data Structures, OOP, Software Engineering
- **Coding Languages** : Java, Python, JavaScript, C++, SQL, HTML/CSS
- **Web Development** : React/Next.js, Node.js, Three.js, Firebase, REST APIs
- **UX/UI Design** : Principles of UX design, Layouts, Transitions (Certified)
- **Development Tools** : Git/GitHub, Docker, Linux, Azure, VS Code
- **Data Science & AI/ML** : Pandas, NumPy, Scikit-learn, TensorFlow, Data Visualization
- **Cybersecurity** : Network Security, Cryptography (learning), Ethical Hacking
- **Business Development** : App marketing, Business funding strategies (Certified)

Professional Skills

- Continuous learning and skill development (demonstrated by FNB certification)
- Critical thinking and complex problem solving
- Project management and team leadership
- Effective collaboration and communication (GitHub Collaboration certified)
- Analytical approach to technical challenges

Key Projects

Campus Dining Services App (React, Node.js, Firebase)

Full-featured application for campus dining with real-time menus and reservations

[Live Demo](#) — [GitHub](#)

Nitro Assault Racing Game (Three.js, JavaScript)

Time-trial racing game with physics and multiple camera perspectives

[Play Demo](#) — [GitHub](#)

Constitutional Archive Management System (Vanilla JS, Node.js, Azure)

Full-stack web application for managing historical constitutional data with NLP search

[Live Demo](#) — [GitHub](#)